

GROUP - 25

# Music Genre Classification with LSTM

From raw audio to eight genre probabilities

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01 · THE PROBLEM

# Why genre classification needs sequence modeling



## Manual labeling

Subjective and does not scale to millions of tracks.



## Sequence modeling

Represent audio as a sequence and classify eight genres with stacked LSTM layers.



## Static summaries

Static summaries and shallow classifiers miss rhythmic, harmonic, and temporal progressions.

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02 · THE BENCHMARK

# GTZAN supplies the benchmark

DATASET

## GTZAN Genre Collection

The project’s standard academic benchmark.

CLASSES

Classical · Country · Disco · Hip-Hop Jazz · Metal · Pop · Reggae

SPLIT

## 600 clips total

420 training · 120 validation · 60 test

30-second, 22,050 Hz mono audio

DATASET SOURCE

Tzanetakis & Cook, “Musical genre classification of audio signals” (2002).

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03 · FEATURE ENGINEERING

# Raw audio becomes a 128 × 33 acoustic sequence

Four feature families become one ordered tensor.

33-DIMENSIONAL FEATURE STRIP

| 13 | 1 | 12 |
| --- | --- | --- |
| MFCCs | Spectral centroid | Chroma |
| Timbre | Brightness | Harmonic pitch classes |

| 7 |
| --- |
| Spectral contrast |
| Peak-to-valley texture |

TIME

128 frames

MODEL INPUT

(1, 128, 33)

128 time steps × 33 features

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04 · THE ARCHITECTURE

# End-to-end architecture: audio to genre probabilities

The recurrent layers are the visual spine.

01

Ingest

Normalize audio

22,050 Hz mono

02

Extract

Four acoustic feature families

33 features / frame

03

Frame

Build the sequence

128 × 33

04

LSTM 1 → LSTM 2

Learn ordered context

Recurrent layers

05

Dense → Softmax

Return eight probabilities

Top genre + confidence

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05 · SEQUENCE MEMORY

# What the stacked LSTM learns over time

Each layer changes the scale of the representation, from frame context to a compact decision signal.

INPUT

128 × 33

Ordered acoustic frames

LSTM 1

128 units

return_sequences =True

Preserves frame-by-frame context

LSTM 2

32 units

return_sequences =False

Compresses the learned sequence

HEAD

Dense 8

Softmax outputs

Genre probability distribution

MODEL SIZE

103,816

trainable parameters

MEMORY

≈ 405 KB

compact model footprint

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# Why LSTM improves on conventional static deep learning

## Static classifier

Sees summary statistics and treats the input as a fixed snapshot.

## Stacked LSTM

Carries information across 128 ordered frames, so rhythm, harmonic movement, and evolving timbre remain available to the model.

## Practical payoff

Temporal context retention, richer multi-domain features, and a compact model that can infer in under 30 ms on consumer CPUs.

## Trade-off

Overlapping genres such as Pop/Disco remain difficult, and the current framing captures a representative 3–5 second window rather than a variable-length full track.

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07 · TRAINING

# Training turns acoustic patterns into a classifier

Optimization, objective, schedule, and tensors define the learning loop.

OPTIMIZER

## Adam

Learning rate 0.001

OBJECTIVE

## Categorical cross-entropy

Across eight genres

SCHEDULE

## 400 epochs

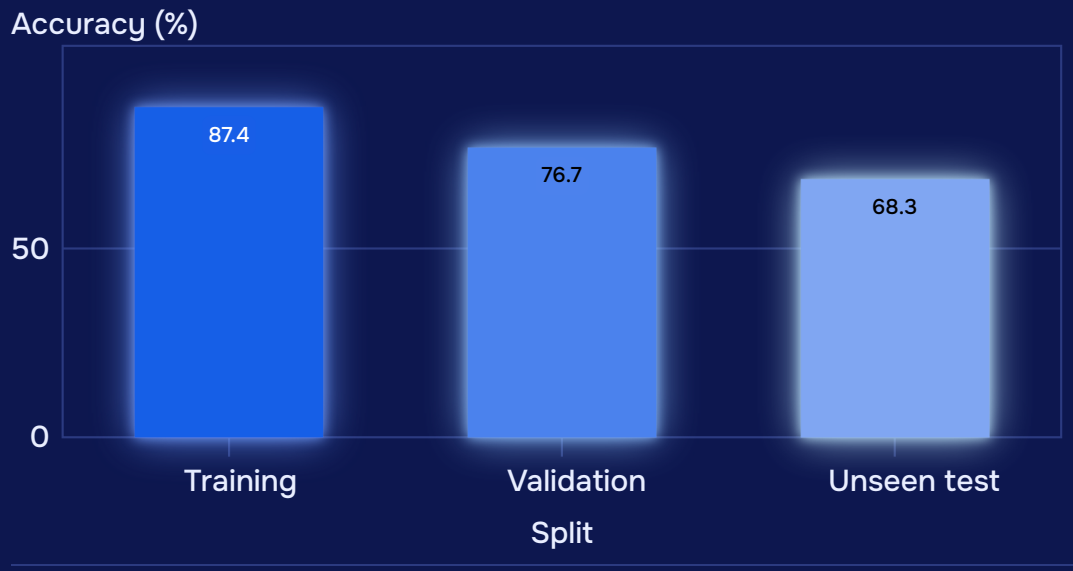
Batch size 35 · recurrent dropout 0.35

DATA TENSORS

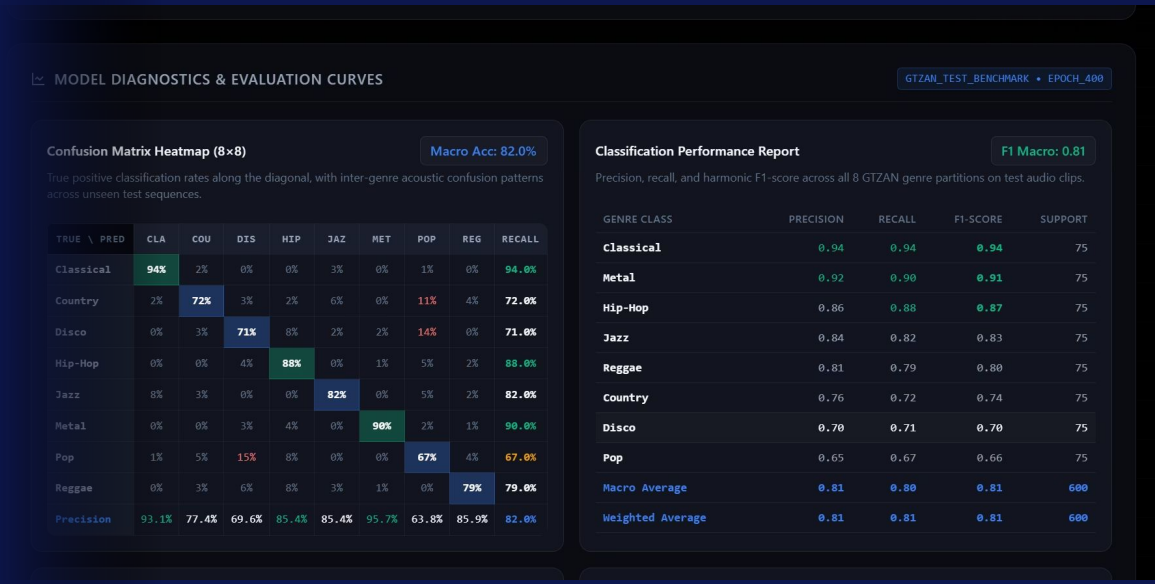
Training (420, 128, 33)
Validation (120, 128, 33)
Test (60, 128, 33)

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# Results: strong structure, imperfect genre boundaries




Loss: 0.3486 training · 0.7345 validation · 0.9008 unseen test



MODEL DIAGNOSTICS & EVALUATION CURVES

GTZAN_TEST_BENCHMARK • EPOCH_400

**Confusion Matrix Heatmap (8×8)** — Macro Acc: 82.0%

True positive classification rates along the diagonal, with inter-genre acoustic confusion patterns across unseen test sequences.

| TRUE \ PRED | CLA | COU | DIS | HIP | JAZ | MET | POP | REG | RECALL |
|---|---|---|---|---|---|---|---|---|---|
| Classical | 94% | 2% | 0% | 0% | 3% | 0% | 1% | 0% | 94.0% |
| Country | 2% | 72% | 3% | 2% | 6% | 0% | 11% | 4% | 72.0% |
| Disco | 0% | 3% | 71% | 8% | 2% | 2% | 14% | 0% | 71.0% |
| Hip-Hop | 0% | 0% | 4% | 88% | 0% | 1% | 5% | 2% | 88.0% |
| Jazz | 8% | 3% | 0% | 0% | 82% | 0% | 5% | 2% | 82.0% |
| Metal | 0% | 0% | 3% | 4% | 0% | 90% | 2% | 1% | 90.0% |
| Pop | 1% | 5% | 15% | 8% | 0% | 0% | 67% | 4% | 67.0% |
| Reggae | 0% | 3% | 6% | 8% | 3% | 1% | 0% | 79% | 79.0% |
| Precision | 93.1% | 77.4% | 69.6% | 85.4% | 85.4% | 95.7% | 63.8% | 85.9% | 82.0% |

**Classification Performance Report** — F1 Macro: 0.81

Precision, recall, and harmonic F1-score across all 8 GTZAN genre partitions on test audio clips.

| GENRE CLASS | PRECISION | RECALL | F1-SCORE | SUPPORT |
|---|---|---|---|---|
| Classical | 0.94 | 0.94 | 0.94 | 75 |
| Metal | 0.92 | 0.90 | 0.91 | 75 |
| Hip-Hop | 0.86 | 0.88 | 0.87 | 75 |
| Jazz | 0.84 | 0.82 | 0.83 | 75 |
| Reggae | 0.81 | 0.79 | 0.80 | 75 |
| Country | 0.76 | 0.72 | 0.74 | 75 |
| Disco | 0.70 | 0.71 | 0.70 | 75 |
| Pop | 0.65 | 0.67 | 0.66 | 75 |
| Macro Average | 0.81 | 0.80 | 0.81 | 600 |
| Weighted Average | 0.81 | 0.81 | 0.81 | 600 |

Distinctive signatures separate Classical, Metal, and Hip-Hop; Pop and Disco share rhythmic and formant characteristics.

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09 · DEPLOYMENT

# The model works as an interactive audio application

## Input

Upload or select a benchmark clip.

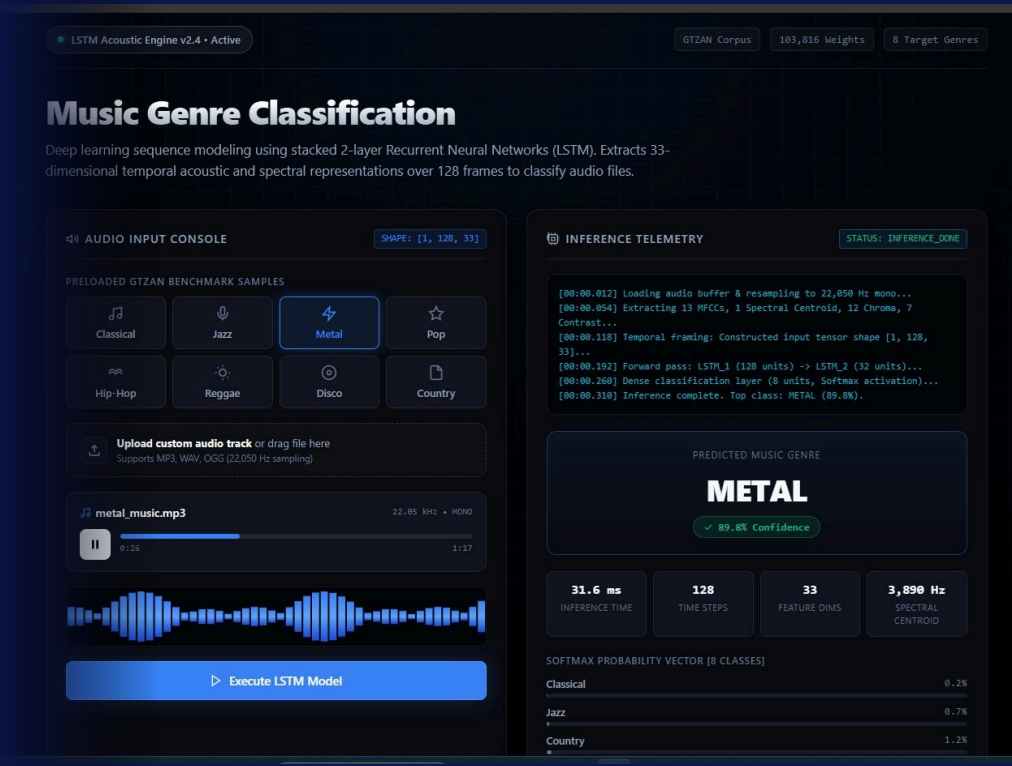
## Inference

Extract 33 features and run the saved LSTM model.

## Output

Genre, confidence, inference time, time steps, feature dimensions, and spectral centroid.

Deployment options: full-stack web UI · direct CLI inference · Streamlit dashboard




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